# IoT-Based Intrusion Detection System

## Using Machine Learning

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# The Problem

### ⚠️ Generalisation Gap

Models trained on one network dataset fail catastrophically on others.
A CICIDS2017+Bot-IoT model dropped from 97% → 36% accuracy on unseen UNSW-NB15 — F1 = 0.00.

### 🔌 Edge Deployment Gap

Even accurate IDS models are too large and slow for IoT gateways.
Raspberry Pi-class devices cannot run multi-hundred-MB ML models in real time.

**97% → 36% accuracy drop on unseen data | 64 percentage-point generalisation gap | F1 = 0.00**

# Objectives

1

### Establish Baseline

Train on CICIDS2017 + Bot-IoT and quantify cross-dataset accuracy degradation

2

### Design SFAF

Map 3 heterogeneous datasets to 12 unified features (85% reduction from 78)

3

### Unified Model

Train one XGBoost model across all datasets; close the gap from 64pp to ~6–8pp

4

### Model Comparison

Benchmark XGBoost vs Random Forest vs LightGBM to justify selection

5

### Edge Compression

Export to ONNX < 50 KB maintaining ≥ 95% accuracy

6

### Latency Benchmarking

Sub-ms inference on Apple Silicon; project Raspberry Pi 4 deployment

# Datasets & Attack Distribution

## CICIDS2017

Canadian Inst. for Cybersecurity

**2.83M records** · 78 features

*DoS, DDoS, Brute Force, Web Attack, Botnet*

## UNSW-NB15

Univ. of New South Wales

**257K records** · 49 features

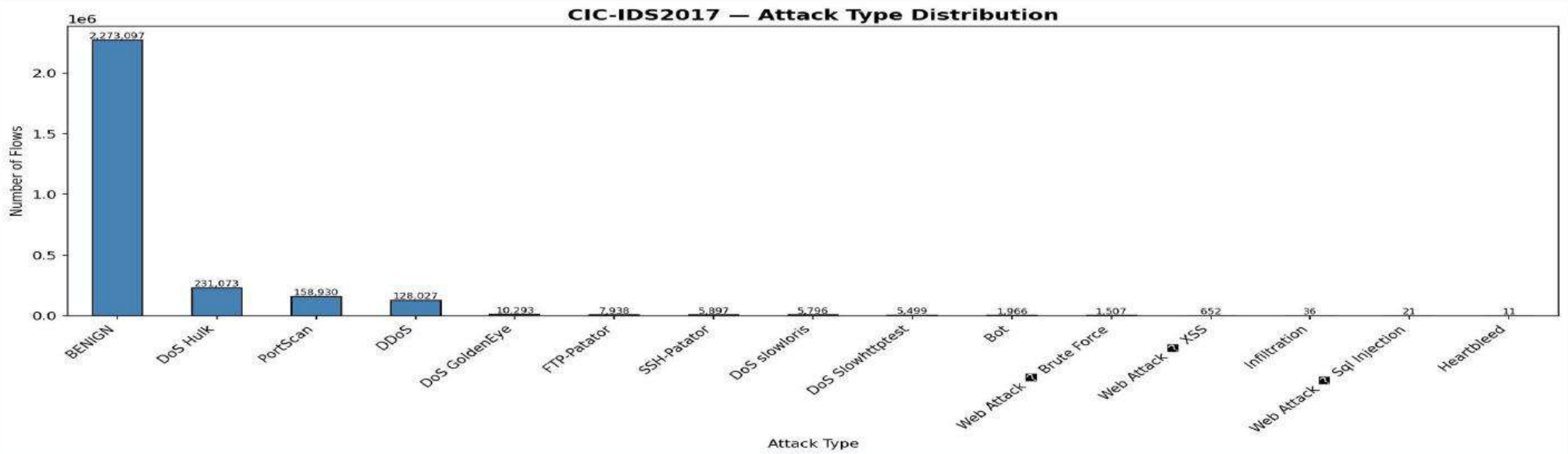
*Fuzzers, DoS, Exploits, Reconnaissance, Shellcode*

## TON-IoT

Univ. of New South Wales (IoT)

**211K records** · 44 features

*Backdoor, DDoS, DoS, MITM, Ransomware, XSS*



# Semantic Feature Alignment Framework (SFAF)

## The Core Innovation

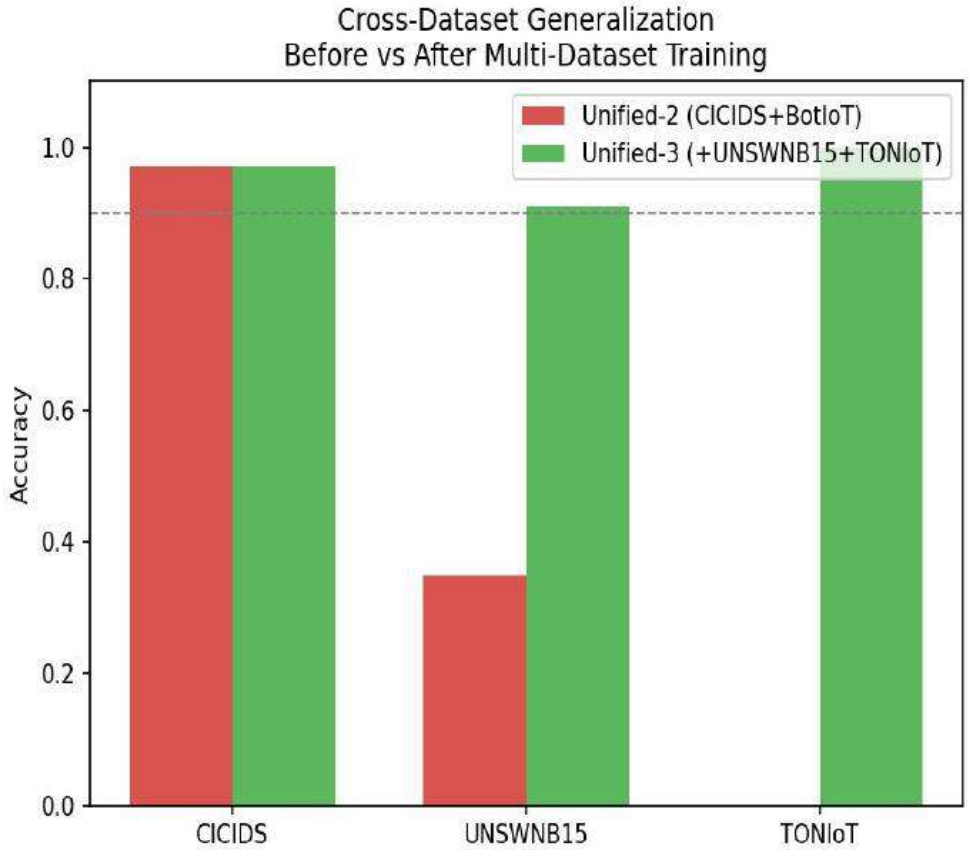
- Different datasets use different column names for semantically equivalent flow properties
- Naive concatenation → 150+ columns, mostly NaN → unified training impossible
- SFAF applies domain-knowledge-driven proxy mapping to a common 12-feature space
- 85% feature reduction (78 → 12 features) concentrating cross-dataset discriminative signal
- All 12 features have positive permutation importance — none are redundant

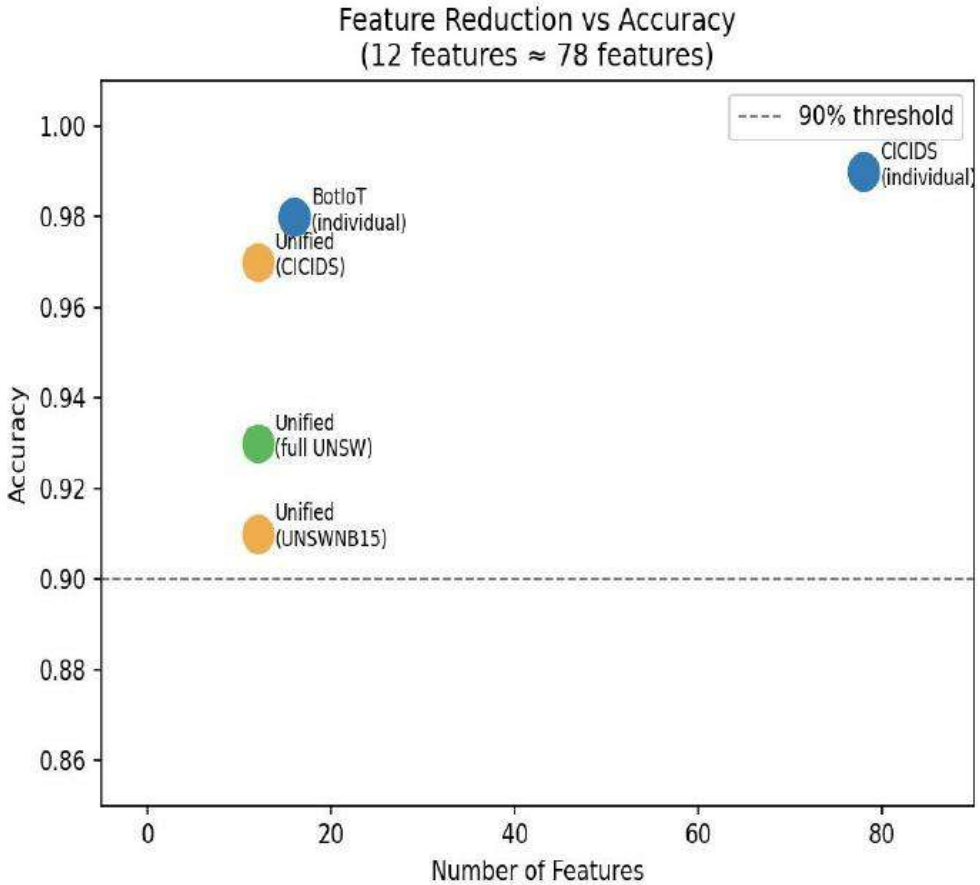
## 12 Unified Features

| | |
|---|---|
| **Flow Duration** | Fwd Packets/s |
| **Total Fwd Packets** | Bwd Packets/s |
| **Total Backward Packets** | Min Packet Length |
| Total Length of Fwd Packets | Max Packet Length |
| Total Length of Bwd Packets | Packet Length Mean |
| Flow Packets/s | Packet Length Std |

*Top 3 account for 52% of total permutation importance*

# The Generalisation Gap — Before vs After SFAF




**Baseline UNSW-NB15 (unseen): 36.1% → Unified SFAF UNSW-NB15: 92.7% → 56.6 percentage-point improvement**

# Training & Evaluation Pipeline

**Dataset Loading**

CICIDS2017
UNSW-NB15
TON-IoT

**SFAF Alignment**

12 unified features

**Preprocessing**

Scale, clean, train/test split

**XGBoost Training**

100 trees
depth=6, lr=0.1

**ONNX Export**

Compress & deply edge model

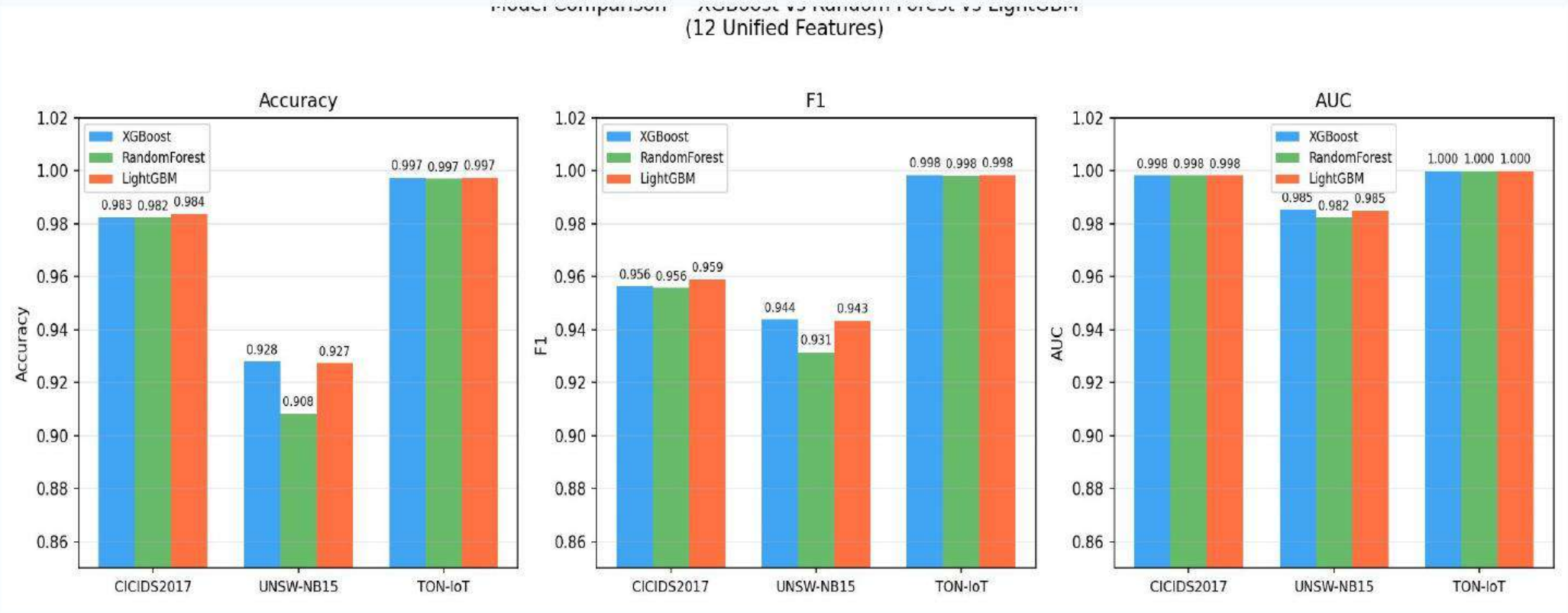
### XGBoost Configuration

```
n_estimators = 100     max_depth = 6
learning_rate = 0.1    tree_method = 'hist'
eval_metric = 'logloss'    random_state = 42
Train/test split: 80 / 20 stratified
```

### Why XGBoost?

- State-of-the-art on tabular traffic data
- 17.6× faster training than Random Forest
- Compact ONNX serialisation (45.8 KB)
- Built-in interpretable feature importance

# Multi-Model Comparison (12 Unified Features)

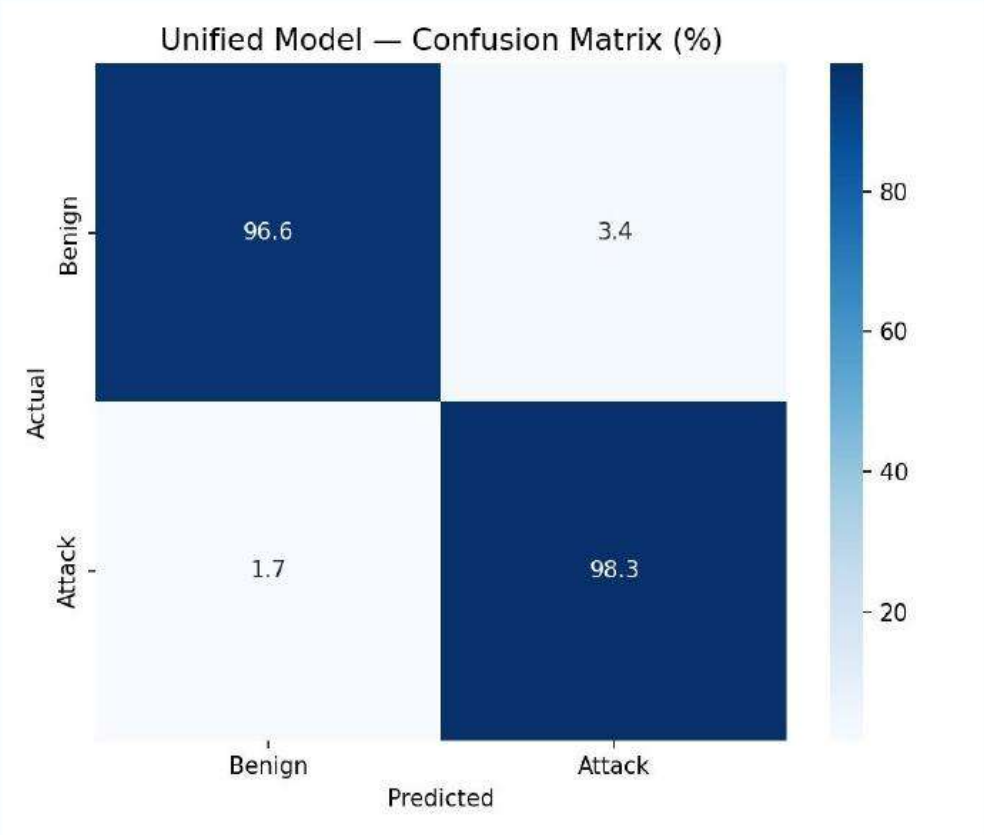


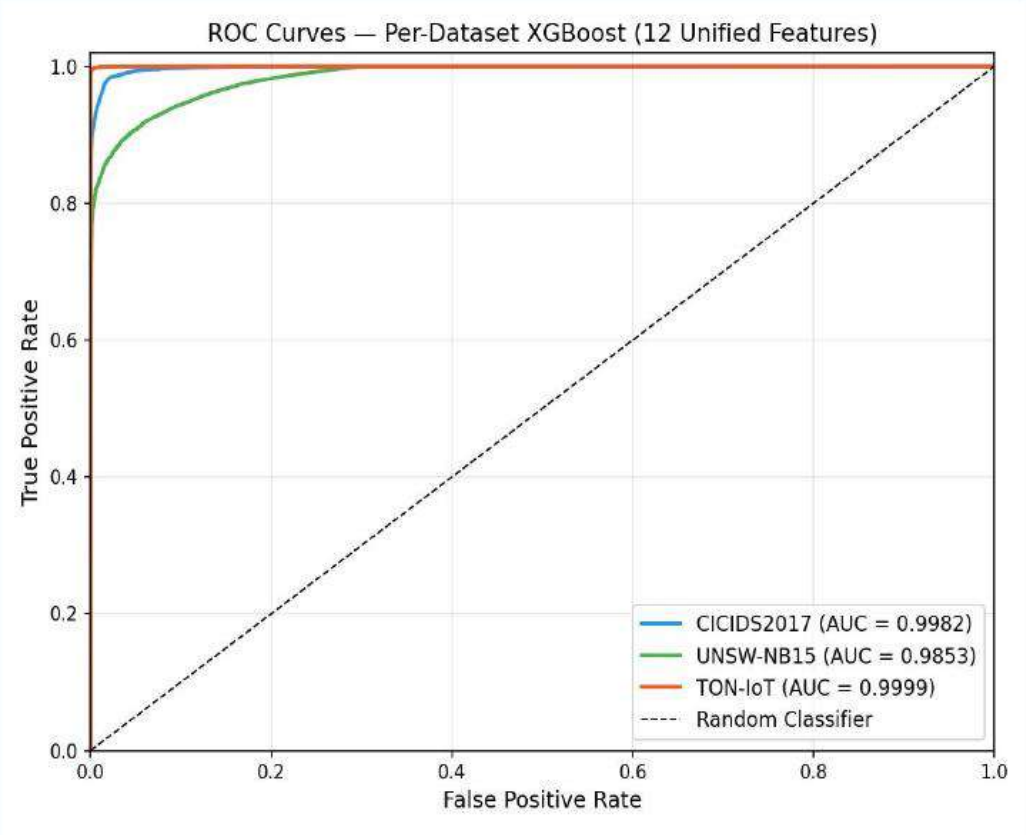

**XGBoost:** Best on UNSW-NB15 (+2% vs RF)

**Speed:** 17.6× faster than Random Forest

**LightGBM:** Comparable accuracy, but larger ONNX

# Unified Model Performance



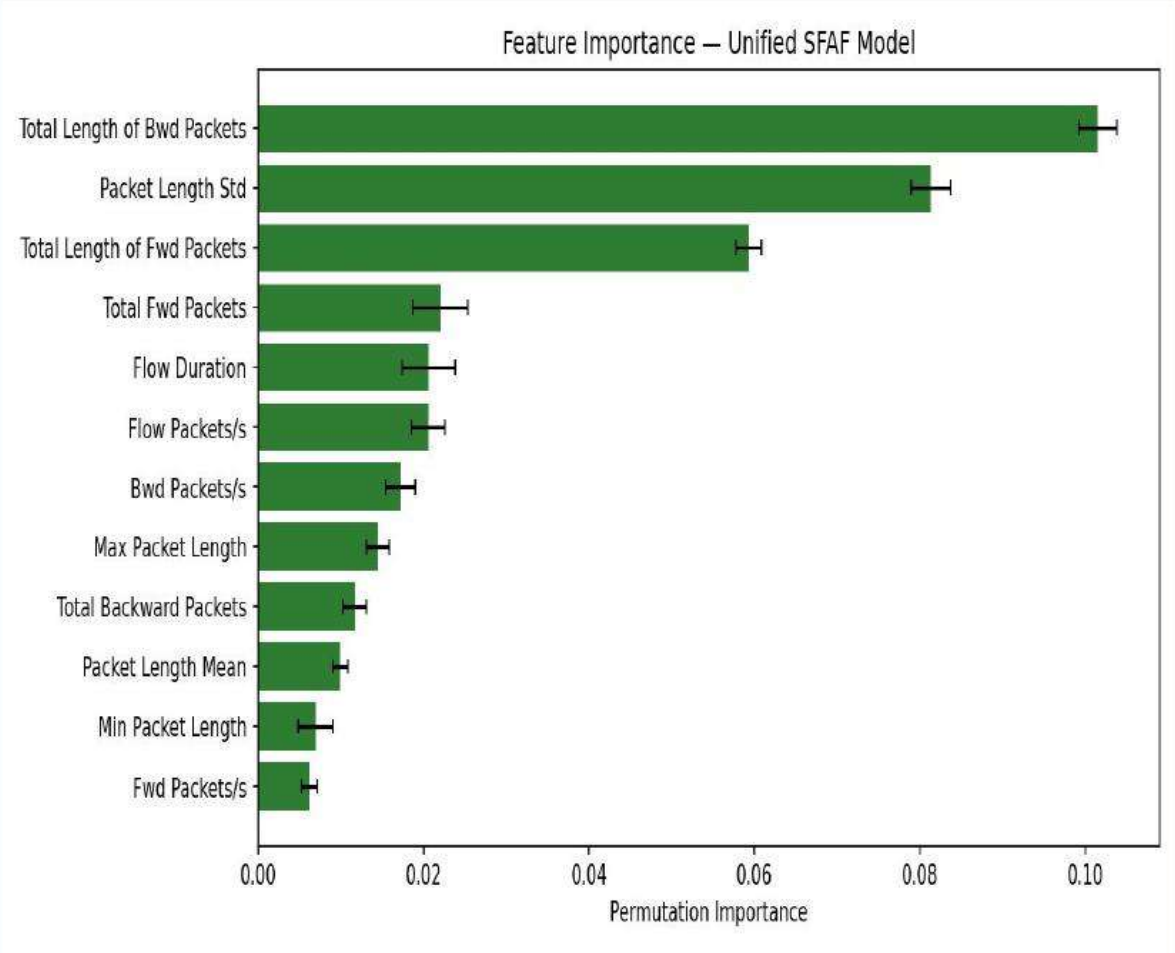




**CICIDS2017** Acc 98.3% AUC 0.9982

**UNSW-NB15** Acc 92.7% AUC 0.9853

**TON-IoT** Acc 99.4% AUC 0.9999

# Feature Importance Analysis




## Key Insights

### Total Length of Bwd Packets

Importance: 0.1019 — single top feature. Attack traffic has anomalously high (exfiltration) or near-zero (DoS) backward byte volumes.

### Packet Length Std

Importance: 0.0658 — high variance distinguishes floods and scans from normal flows.

### Top 3 features

Account for 52% of total permutation importance across all datasets.

### All 12 features

Positive importance > 0 — confirms zero redundancy in SFAF design.